

Computer Reliability



Computer as a reliable machine

- The computer is a reliable machine which is responsible for performing all the tasks and operations at high speed and almost 100% accuracy.
- The computers software and hardware works according to the request or demand given to them by users.
- There are several attributes that we should check while buying, making, and using a computer.
- People who sell computers often talk about the product's reliability, since reliable products contain no technical errors.

Reliability of computer system

- The computer performs all activities for which it is designed and developed with great accuracy this makes them very reliable machines. This is considered as the main reliability of computer system.
- Reliability is **an attribute of any computer-related component (software, or hardware, or a network, for example) that consistently performs according to its specifications.**
- Computer reliability in its simplest terms means whether the computer will be available when you need it.

Reliability Characteristics of Computer

- When computing mathematical calculations, a computer works much faster and accurately and can perform millions of instructions per second.
- Although the computer performs calculations with 100% accuracy, **errors may occasionally occur** due to data inconsistency or inaccuracy.
- **A computer is reliable as it gives consistent results** for a similar set of data; when commanded for similar set of instructions multiple times.
- The results of computer entirely depend upon the instructions given to them.

Benefits and Advantages of Reliability in Computer

- A business reputation is over dependent on the reliability of its computer system.
- We can expect more accurate results compared to humans.
- All tasks and operations are performed at high speed therefore jobs and tasks are completed at given time which builds good reputations in client's eyes.
- The modern computer can store enormous amount of data almost permanently that makes us free from worries and this considered as the major advantages of reliability in computer system
- The accountings are done with the help of computer system due to their reliability.

Importance of Network Reliability



- In computer networks the data and instructions are transferred from one computer to another with speed.
- The computer networks are good source of data communications.
- The *internet* is considered as the best source of data communications which is cheap and fast.
- The networks plays and important role in data and information sharing around the globe which saves money and time.
- These helps in great deal of consistency and reliability.
- Voice call or video conferencing is made simple in networking that connects and helps in easy communication.

Why is Computer Reliability Important?

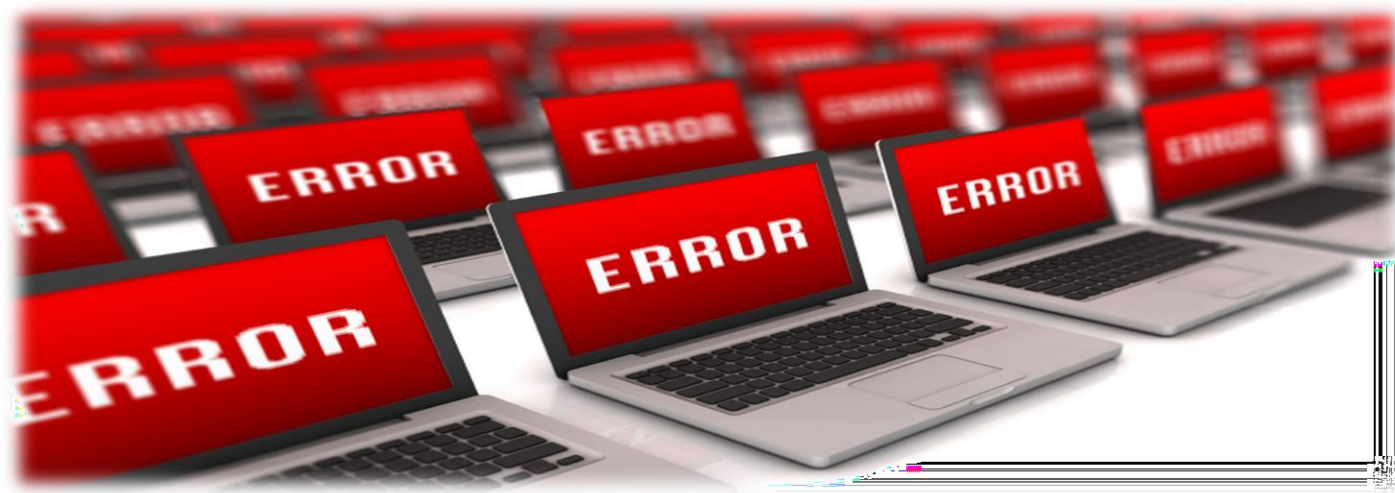
- Reliability is how well the computer system performs; they require functions that it has been designed for and do them form without failure.
- This is important because if the system is not reliable, then the system may not function as it was supposed to do when the system fails, then they cause a massive amount of damage.



Reliability

Data-Entry or Data-Retrieval Errors

- A computerized system may fail because wrong data entered into it
- A computerized system may fail because people incorrectly interpret data they retrieve



Computer failure causes:

- **Faulty design**
- **Careless or insufficiently trained users**
- **Poor user interfaces**
- **Hardware/Software malfunctions**
- **Scope/Application inconsistency**



Computer users:

- Should understand limitations of the computers
- Need for proper training
- Need for responsible use
- Difference between good products and bad ones
- **Computer Professional should:**
 - Study computer failures (reasons)
 - Study computer ethics



Three Categories of Failures



1. Problems for individuals
2. System failures that affect large numbers of people or cost large amounts of money
3. Problems in safety-critical applications

Problems for Individuals

- Errors is design and/or implementation of programs
- Not enough care
- Not enough testing - reasonable range
- Not enough training



Database Accuracy Problems

- Information in database is not accurate
- Automatic entering of info - mistakes can be overlooked
- Copies of incorrect info can be sent to other systems
- Not knowledgeable enough about the system

Causes

- Large population
- Most of our financial interactions are with strangers
- Automated processing without human common sense
- Overconfidence in accuracy of data
- Lack of accountability/responsibility

Consumer Hardware and Software

- Usually have more serious errors in their first releases
- Regularly sold with known bugs/virus
- Hardware also has flaws
- Tradeoff between cost, debugging, and marketing
- Dishonesty, denials of problems, lack of adequate response to complaints

System Failures

- Lots of money \$\$\$\$
- Complete shutdown of basic services
- Areas:
 - Communications related companies
 - Business and financial systems
 - Military



WHY?

- Not enough testing
- Technical difficulties
- Poor management decisions
- Dishonesty in promoting the system and responding to problems



Communications

- Phone Service providers (PTA, Ufone..)
- How Bad?
 - phone calls
 - Communications for airports
 - cellular phones (Samsung, apple)



Business and financial systems



- Incorrect information affects business
- Loss of data
- Stock exchange companies
- ATM
- Data accuracy (customers. Employees)



Military

- Data management
- Weapons system design
- Battle simulation
- Battle management
 - command/control
 - communications
 - intelligence
- Nuclear war



Safety Critical Applications

- Use of computers is increasing rapidly in these areas
- Use of computers in these areas can save \$

- Areas

- Military
- Power plants
- Aircraft
- Trains
- Automobiles

Medical Applications

Safety-critical systems are those systems whose failure could result in loss of life, significant property damage or damage to the environment. There are many well-known examples in application areas such as **medical devices, aircraft flight control, weapons and nuclear systems.**

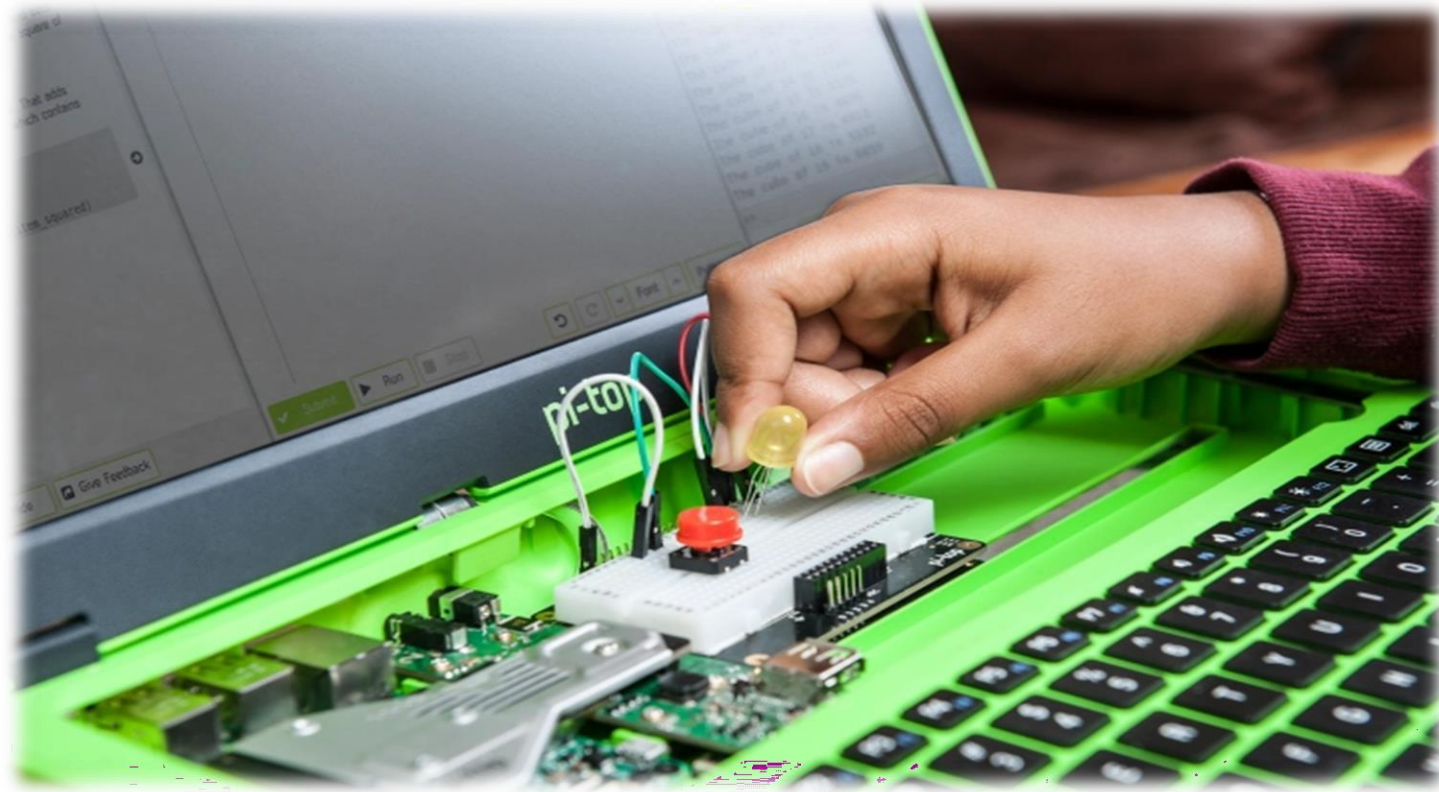
Ways to prevent problems

- Good computer systems
- Good training
- Accountability
- Individual responsibility
- Management responsibility
- IEEE/ACM Code of Ethics



Professional Techniques

- Use good software engineering techniques at all stages of development:
 - Requirements
 - Specifications
 - Design
 - Implementation
 - Documentation
 - Testing



Professional Techniques

- Study the techniques and tools available
- Knowing or learning enough about the application field and the software or systems being used.



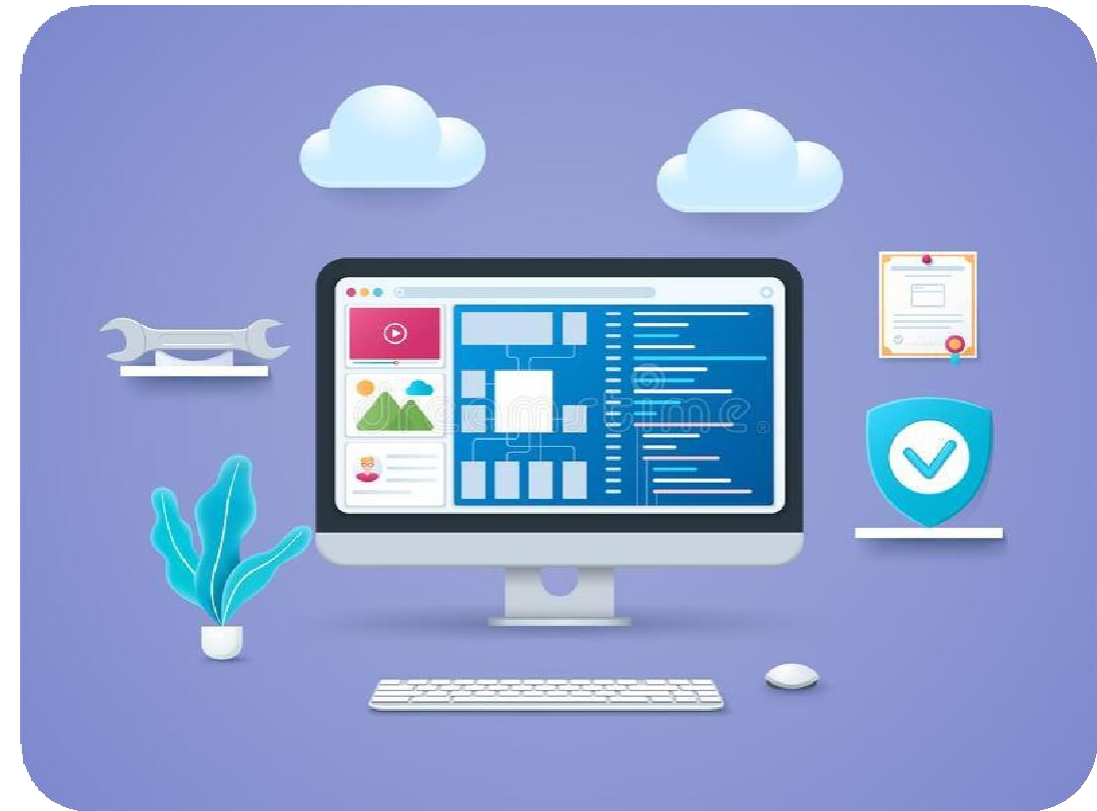
Laws and Regulations

- Criminal and Civil penalties
- Suits against company that designs or sells the system
- Criminal charges when fraud or criminal negligence occurs
- Need contracts
- Need well designed laws and standards



Professional licensing

- Licensing of software development professionals to protect against poor quality and unethical behavior
 - Specific training
 - Passing competency exam
 - Ethical requirements
 - Continuing education

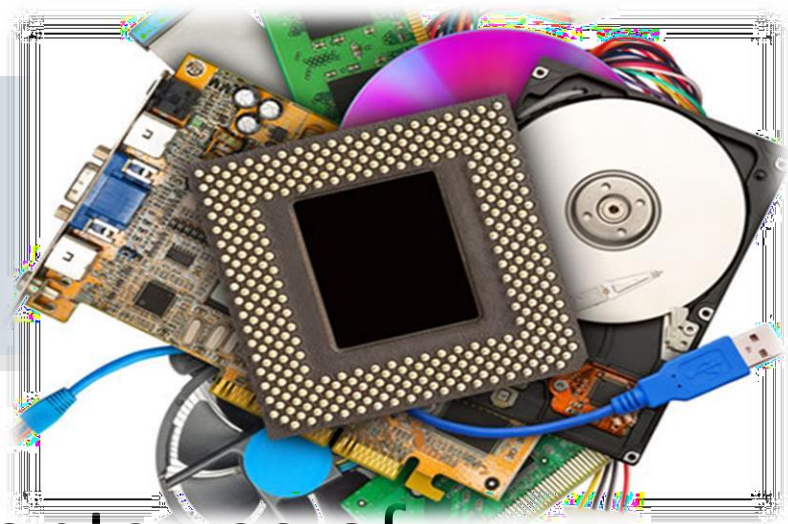


What is Reliability of Software?

- Assume data correctly fed into computerized system.
- System may still fail if there is an error in its programming (a bug)
- The **System software** are the most important aspect of computer system, operating system and device drivers falls in the category of system software.
- The operating system once installed can help in working with other application software's like MS-word, MS-Excel etc.

- The **operating system (OS)** is reliable software which can be considered as backbone of any computer system.
- **Application software** are installed and used when OS is installed in computer. The application software is used primarily for accounting and tracking complete records of business.
- Majority of the business owner rely heavily on the accounting software for accurate results this helps in increasing reliability of software's.

Software Errors - bugs



- Fixes are complex, expensive, and prevents use of machine while fixing
- Bugs
 - can be hard to detect
 - importance of self checking
 - importance of using good programming techniques

Moral Responsibility of Software Manufacturers

- If vendors were responsible for harmful consequences of defects
 - Companies would test software more
 - They would purchase liability insurance
 - Software would cost more
 - Start-ups would be affected more than big companies and thus
 - Less innovation in software industry
 - Software would be more reliable
- Making vendors responsible for harmful consequences of defects may be wrong
- Consumers should not have to pay for bug fixes

